COMPLETED STUDY / 5 NOVEMBER 2026

Environmental Sound Classification on ESC-50

Architecture, robustness and representation

Final research report with original fitted-task results, completed three-seed robustness, encoder-token interventions, and all 19 scientific figures.

65 original tasks | 60 neural fits | 19 figures

Every current score is development-validation evidence. Historical fold-5 and fold-rotation figures are retained in a separate appendix. No new independent confirmation is claimed.

Manuscript drafted and reviewed through role-based ChatGPT discussions; existing experiment artifacts supplied the results. No additional attention training was required to complete this study.

Abstract

This study investigates environmental sound classification on ESC-50 through three research questions: the contribution of temporal modeling and attention mechanisms (RQ1), robustness to acoustic perturbations (RQ2), and the influence of audio feature representation and sampling configuration (RQ3).

The research comprises 65 experimental tasks, including 60 neural-network training runs, together with additional inference-only investigations using existing checkpoints. A source-strict development split was used to reduce source overlap between training and validation.

Across three training seeds, the baseline CNN achieved 72.61% validation accuracy, while the CNN-BiGRU model achieved 73.87%. Channel Attention produced the highest observed architecture-group mean accuracy of 74.37%, although its advantage over BiGRU was only 0.50 percentage points (pp), with a source-bootstrap confidence interval spanning zero.

Noise-augmented CNN training produced substantially stronger robustness under matched Gaussian noise. At 10 dB signal-to-noise ratio (SNR), accuracy increased from 23.37% for the baseline CNN to 61.31% for the noise-trained CNN.

Feature experiments showed consistent advantages for 32-band over 64-band Log-Mel representations within a separate frequency-pooled CNN architecture. Temporal interventions further demonstrated that BiGRU predictions were sensitive to encoder-token ordering, whereas CNN predictions were invariant under the tested permutations.

These findings support architecture-specific conclusions about temporal aggregation, noise robustness, and feature configuration. However, repeated development-set exposure, limited training seeds, and the absence of independent confirmation restrict broader generalization claims.

1. Research Questions

This study addresses three questions.

RQ1 - Architecture and Temporal Modeling: Does recurrent temporal aggregation improve environmental sound classification, and do attention mechanisms provide additional benefits? What does sensitivity to encoder-token ordering reveal about the models?

RQ2 - Acoustic Robustness: How do baseline, recurrent, and noise-trained architectures respond to additive Gaussian noise and amplitude variation? Does noise-augmented training reduce degradation under matched perturbations?

RQ3 - Audio Representation: How do sampling rate, frequency coverage, and feature dimensionality affect classification performance under a controlled architecture and training budget?

The objective is to identify supported empirical relationships rather than assume that greater architectural complexity necessarily improves recognition accuracy.

2. Data and Methods

2.1 Dataset and Validation Protocol

ESC-50 contains 2,000 five-second environmental audio recordings distributed across 50 classes. The original recordings are provided at 44.1 kHz.

The primary experiments used the following source-strict development split:

Partition	Configuration
Training	Folds 1-3; 1,200 recordings
Validation	Fold 4; 398 recordings from 295 `src_file` groups
Designated test	Fold 5; 400 recordings
Validation selection	Minimum validation cross-entropy
Training repetitions	Seeds 101, 202, and 303

Source identity was considered when constructing the development validation set to reduce source-level leakage.

However, all official folds had been exposed during the broader project history. Consequently, the designated fifth fold cannot be described as an untouched blind test set, and the reported results are development-validation findings.

2.2 Audio Processing

The principal audio preprocessing pipeline was:

WAV → Resampling → Log-Mel Spectrogram → Normalization

The main configuration used:

• Sampling rate: 16 kHz

• FFT size: 512

• Hop length: 256

• Mel bands: 64

• Logarithm: Natural logarithm

• Feature tensor shape: [1, 64, 313]

Normalization statistics were estimated using training data only and subsequently applied to validation inputs.

2.3 Experimental Design

The original experimental matrix contained 65 tasks:

Experiment group	Tasks
Classical baselines	2
Architecture comparisons	15
Augmentation experiments	9
Feature and sampling experiments	36
Original robustness stress tasks	3
Total	**65**

The 60 neural-network training runs used 60 epochs, batch size 32, AdamW, initial learning rate 0.001, weight decay 0.0001, and cosine learning-rate scheduling.

For each run, the checkpoint with the lowest validation cross-entropy was selected.

Additional RQ1 and RQ2 investigations reused existing checkpoints for inference without retraining.

Reported architecture results are means across three fixed training seeds, accompanied by sample standard deviations. Paired confidence intervals were estimated using 4,000 source-level bootstrap resamples over the fixed fitted models. These intervals do not characterize the population of possible training seeds. Each source resample retains every constituent clip and uses its actual, varying clip denominator; the same source draws are shared across paired models and fixed seeds. The paired-analysis RNG seed is 20261104; the temporal-intervention RNG seed is 20261105.

3. Results

3.0 Classical Baselines

Method	Accuracy	Macro-F1
Log-Mel statistics + logistic regression	42.21%	38.86%
DCT cepstral statistics + linear SVM	40.20%	37.50%

The feature families and selection procedures differ; the score difference is not a classifier-only comparison.

3.1 RQ1 - Architecture, Attention, and Temporal Modeling

Architecture Comparison

Model	Parameters	Accuracy, mean ± SD	Macro-F1, mean ± SD
CNN	49,058	72.61 ± 0.25%	71.39 ± 0.58%
CNN-BiGRU	251,810	73.87 ± 1.26%	72.97 ± 1.36%
Temporal Attention (TA)	251,939	72.28 ± 2.41%	71.52 ± 2.41%
Channel Attention (CA)	252,906	74.37 ± 1.15%	73.48 ± 1.00%
Temporal + Channel Attention	253,035	72.86 ± 1.26%	71.69 ± 1.35%

BiGRU improved mean accuracy over CNN by 1.256 pp, but the paired 95% source-bootstrap interval was [-1.806, 4.167] pp.

Attention comparisons against BiGRU produced:

Comparison	Difference (pp)	95% source-bootstrap CI (pp)
CA - BiGRU	+0.503	[-1.217, 2.167]
TA - BiGRU	-1.591	[-4.473, 1.250]
TA+CA - BiGRU	-1.005	[-3.782, 1.737]

Channel Attention achieved the highest observed mean, but its paired improvement was small: approximately one, three, and two additional correct predictions across the respective seeds.

Neither temporal attention nor combined attention improved mean accuracy over the BiGRU reference.

These findings do not establish a reliable general advantage for attention-based architectural complexity.

Encoder-Token Order Interventions

A separate inference-only investigation examined whether trained models depended on temporal-token ordering.

The intervention directly rearranged the 39 temporal tokens in the encoder output, preserving the embedding associated with each token. It did not reverse the original WAV waveform.

Five conditions were evaluated: identity, reverse, roll by 19 positions, and two fixed shuffles using seeds 17 and 29.

Across six existing checkpoints, this produced 30 prediction sets and 11,940 predictions.

Intervention	BiGRU accuracy	Change from identity (pp)	95% CI (pp)
Identity	73.87%	Reference	-
Reverse	68.84%	-5.03	[-7.54, -2.60]
Roll 19	-	-1.42	[-2.94, 0.09]
Shuffle 17	-	-1.09	[-2.64, 0.42]
Shuffle 29	-	-0.08	[-1.47, 1.32]

The CNN exhibited zero prediction changes across all tested token permutations.

This invariance is consistent with its temporal aggregation structure; it does not imply that the CNN ignores temporal information.

Each encoder token already incorporates approximately 368 milliseconds of local temporal context.

BiGRU sensitivity to reversal therefore demonstrates dependence on the arrangement of learned local representations, not necessarily the recognition of long-range acoustic event direction.

The shuffle conditions also changed numerous individual predictions despite relatively small net accuracy differences, indicating that newly introduced errors and corrected errors partially offset one another.

Possible explanations include event directionality, onset and offset structure, silence positioning, boundary effects, and nonlinear recurrent-state aggregation. The current interventions cannot uniquely distinguish these mechanisms.

3.2 RQ2 - Robustness to Acoustic Perturbations

Noise robustness was evaluated using existing CNN, BiGRU, and noise-trained CNN checkpoints.

After completing the missing inference evaluations, the investigation contained three models, three training seeds, and six conditions, totaling 54 prediction sets.

Gaussian noise was added at specified SNR levels, with the same fixed noise realization shared across models and training seeds for each recording-condition pair.

Post-perturbation RMS-SNR was measured, clipping was avoided, and checkpoint-specific training normalization was retained.

Accuracy Under Gaussian Noise

Evaluation condition	CNN	BiGRU	Noise-trained CNN
Clean	72.61%	73.87%	73.12%
Gaussian, 20 dB	47.74%	50.84%	**70.27%**
Gaussian, 10 dB	23.37%	23.87%	**61.31%**
Gaussian, 0 dB	7.20%	5.86%	**25.38%**

The noise-trained CNN maintained substantially higher accuracy under the evaluated Gaussian perturbations.

Its paired advantages over the baseline CNN were:

Condition	Accuracy advantage (pp)	95% CI (pp)
20 dB	+22.53	[17.35, 27.64]
10 dB	+37.94	[32.08, 43.88]
0 dB	+18.17	[14.55, 21.87]

At 10 dB, after accounting for the clean-condition accuracy advantage, noise training reduced the accuracy loss by 37.44 pp [31.31, 43.74].

By contrast, the ±6 dB amplitude-control conditions provided no clear robustness advantage.

In the augmentation comparison, the clean-validation accuracies were:

Augmentation	Accuracy, mean ± SD
Noise augmentation	73.12 ± 0.87%
SpecAugment	71.86 ± 0.79%
Mixup	71.61 ± 0.66%

The principal RQ2 result is that matched Gaussian-noise augmentation substantially reduced sensitivity to additive Gaussian noise without a large observed reduction in clean accuracy.

However, the result does not demonstrate robustness to real-world environmental noise, non-Gaussian interference, or unseen noise-generation processes.

The reported SNR describes the added perturbation, not the total environmental noise already present in an original recording.

3.3 RQ3 - Feature Representation and Sampling Configuration

RQ3 used a separate frequency-pooled CNN with 26,658 parameters.

Twelve feature-and-sampling configurations were evaluated across three training seeds, yielding 36 training runs. Four representations were crossed with three sampling/bandwidth settings: Log-Mel 64, Log-Mel 32, 40-coefficient DCT cepstral features, and Log-Linear 64. FFT windows and hops were held at 32 ms and 16 ms across sampling rates.

Selected Log-Mel comparisons were:

Sampling rate	Maximum frequency	Mel64 accuracy	Mel32 accuracy	Difference (pp)
16 kHz	8 kHz	51.59%	**55.61%**	+4.02
32 kHz	8 kHz	51.26%	**55.70%**	+4.44
32 kHz	16 kHz	51.09%	**57.12%**	+6.03

The 32-band representation consistently outperformed the corresponding 64-band representation in these comparisons.

This suggests that lower spectral dimensionality may offer a useful inductive bias within the tested architecture, potentially reducing the burden of learning from sparse training data.

Nevertheless, this explanation remains hypothetical. The experiment does not isolate all possible mechanisms behind the observed differences.

Thirty of the 36 runs selected their best checkpoints during epochs 51-60. However, the 32-band validation-loss advantage appeared earlier in training.

Late checkpoint selection alone does not establish that training had failed to converge.

The feature findings apply to the separate pooled-CNN architecture and should not be generalized directly to the principal CNN-BiGRU models.

4. Error Analysis

Persistent errors were investigated across the existing fitted models.

The CNN produced 83 recordings that were incorrectly classified by all three seeds, while the BiGRU produced 64 such recordings.

Across four model families, 41 recordings from 31 source groups were consistently misclassified.

These shared failures suggest that certain validation examples remain difficult across the tested modeling choices.

Four examples were additionally selected based on existing CNN/BiGRU disagreements: clapping, siren, glass breaking, and helicopter.

Their BiGRU correctness counts changed under encoder-token reversal as follows. Each count concerns three fitted models on the same single recording, not three independent recordings:

Example	Identity	Reverse
Clapping	3/3	2/3
Siren	3/3	3/3
Glass breaking	3/3	1/3
Helicopter	1/3	3/3

The selected glass-breaking example contained substantial early energy followed by an extended lower-energy region.

This observation raises possible explanations involving silence positioning, event boundaries, and recurrent-state dynamics.

However, these four examples were deliberately selected from model disagreements and are not representative category-level samples.

No listening-based verification or independent label audit was completed. Prediction CSV files alone cannot establish incorrect ground-truth labels or a specific acoustic failure mechanism.

5. Discussion

5.1 Architectural Complexity and Temporal Information

The modest BiGRU advantage, small Channel Attention gain, and negative mean results for Temporal Attention and combined attention provide no evidence that simply increasing attention complexity guarantees better classification.

The recurrent model is sensitive to the ordering of encoder representations, whereas the pooled CNN is invariant under the tested permutations.

Importantly, sensitivity to temporal-token arrangement is not equivalent to a demonstrated benefit from long-range temporal reasoning.

The encoder already captures local temporal patterns, and recurrent nonlinear aggregation may be affected by token position, silence distribution, and boundary structure.

These competing explanations remain unresolved.

5.2 Robustness and Representation

RQ2 provides the strongest observed performance contrast: noise training substantially improved classification under matched Gaussian disturbances.

This is meaningful within the tested perturbation family but should not be interpreted as verified cross-distribution robustness.

RQ3 shows that higher-dimensional spectral representations do not necessarily provide better accuracy in a data-limited setting.

Together, the findings show that performance varied across the tested representations, architectures, and training or perturbation conditions. The experiments do not identify all causal interactions among these factors.

5.3 Decision on Further Attention Training

All nine attention-specific training runs were completed, representing three variants across three seeds. They form part of the 15 architecture-training runs, not an additional unfinished experiment.

Therefore, the current investigation does not require further attention training or another large training matrix to reach its stated conclusions.

If the objective changes from completing this research to maximizing recognition accuracy, AudioSet-pretrained transfer learning using architectures such as PANNs or AST would be a reasonable separate research direction.

Such a study would introduce additional questions concerning pretraining data, dataset overlap, feature preprocessing, model capacity, and evaluation comparability.

No pretrained weights were downloaded or evaluated in this project, and compatibility, licensing, and data overlap have not been fully verified.

Consequently, no performance improvement is claimed or guaranteed. A pretrained-model comparison would also not isolate attention as the sole explanatory factor.

6. Limitations

Several limitations constrain the interpretation of the findings.

Development-data exposure. All official folds had historical exposure during the project. The reported results are not independently confirmed test-set estimates.

Limited model repetitions. Neural comparisons rely on three fixed training seeds. Source-bootstrap confidence intervals quantify variation over sampled source groups conditional on those fitted models, not variation over independent training runs.

Exploratory multiplicity. Numerous architectures and configurations were evaluated using development data. The reported intervals are not adjusted for this broader exploratory selection process.

Noise generalization. Gaussian training and evaluation used the same broad perturbation family. Each recording-condition pair had a single fixed noise realization. The intervals do not cover uncertainty from newly generated noise or retrained models.

Temporal-mechanism identification. Encoder-token permutations preserve local embeddings and are not equivalent to waveform reversal. Directionality, silence, boundary structure, and recurrent aggregation remain competing explanations.

Architecture-dependent feature findings. RQ3 employed a separate pooled CNN, preventing direct generalization of its Mel-band results to the primary architecture.

Source grouping and near duplicates. Source-strict grouping addresses known recording IDs; it does not rule out perceptual near-duplicates with different IDs. A complete perceptual near-duplicate audit was not performed.

Incomplete independent confirmation. No new blind-test evaluation, real-environment noise validation, pretrained-transfer experiment, or listening-based error-label verification was conducted.

7. Conclusion

This study completed a systematic comparison of environmental sound classification architectures, augmentation strategies, and audio representations on ESC-50.

Three main conclusions emerge.

First, recurrent temporal aggregation modestly improved mean validation accuracy, but the available uncertainty estimates do not establish a reliable general advantage. Channel Attention achieved the highest observed architecture-group mean, while additional temporal attention did not consistently improve performance. Encoder-token interventions demonstrated order sensitivity in BiGRU without uniquely identifying its underlying acoustic mechanism.

Second, noise-trained CNNs substantially outperformed standard CNNs under matched Gaussian perturbations, especially at 10 dB SNR. This supports the effectiveness of targeted noise augmentation within the evaluated distribution, not universal environmental robustness.

Third, in three tested configurations, Log-Mel32 achieved higher mean validation accuracy than Log-Mel64 within the pooled-CNN architecture; the underlying mechanism remains unidentified.

The existing evidence is sufficient to complete the current research without further attention training or a new model-training campaign.

Future pretrained-transfer investigations may pursue higher recognition accuracy, but they represent a distinct objective requiring separate controls and independent evaluation.

Performance varied across the tested representations, architectures, and perturbation conditions; their underlying causal interactions remain unresolved.

Appendix A. Development-study and follow-up figure atlas

These figures retain their original evaluation scopes. In particular, the original robustness chart uses seed101 only; the completed three-seed result is shown separately. SD across fits is not a dataset confidence interval.

Figure 1. Three-seed development validation; SD is not a confidence interval. - full-page figure atlas.

Figure 2. Augmentation affects training only; validation recordings stay clean. - full-page figure atlas.

Figure 3. Within-seed deltas; these are not significance tests. - full-page figure atlas.

Figure 4. Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track. - full-page figure atlas.

Figure 5. Original seed101 robustness comparison: one fitted checkpoint per model - full-page figure atlas.

Figure 6. Fixed-seed101 validation curves; checkpoints use minimum cross-entropy. - full-page figure atlas.

Figure 7. A real validation waveform and Log-Mel representation - full-page figure atlas.

Figure 8. Completed three-seed Gaussian-noise and gain results; bars show sample SD, not confidence intervals - full-page figure atlas.

Figure 9. Feature-study loss trajectories: Log-Mel 32 and Log-Mel 64, three fixed seeds - full-page figure atlas.

Figure 10. Four selected real-audio error cases: waveform and Log-Mel evidence, not causal explanations - full-page figure atlas.

Figure 11. CRNN encoder-token intervention effects with conditional source-bootstrap intervals - full-page figure atlas.

Appendix B. Historical baseline figures: separate evaluation scope

The following eight figures belong to the earlier CNN/CRNN study, including historical fold-5 evaluation and fixed-recipe fold rotation. They are retained for completeness and provenance, not presented as new independent confirmation or merged into the current development averages.

Figure 12. Historical study - Model comparison - full-page figure atlas.

Figure 13. Historical study - Learning curves - full-page figure atlas.

Figure 14. Historical study - Named confusions - full-page figure atlas.

Figure 15. Historical study - Order probe - full-page figure atlas.

Figure 16. Historical study - Synthetic stress - full-page figure atlas.

Figure 17. Historical study - Five-fold characterization - full-page figure atlas.

Figure 18. Historical study - CNN primary fold-5 confusion matrix - full-page figure atlas.

Figure 19. Historical study - CRNN (CNN + BiGRU) primary fold-5 confusion matrix - full-page figure atlas.

Appendix C. Original fitted-task ledger

The original matrix has 62 fitted tasks (60 neural fits and two classical fits) plus three original stress tasks. The following are the original development-validation records; follow-up inference adds no fitted models.

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
A_logistic	42.21	38.86	not applicable
A_cepstral_svm	40.20	37.50	not applicable
B_cnn_freq_s101	72.61	71.94	47

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
B_cnn_freq_s202	72.36	70.79	49
B_cnn_freq_s303	72.86	71.45	46
B_crnn_freq_s101	75.13	74.21	58
B_crnn_freq_s202	73.87	73.18	58
B_crnn_freq_s303	72.61	71.52	30
B_crnn_freq_ta_s101	74.87	74.02	46
B_crnn_freq_ta_s202	70.10	69.22	58
B_crnn_freq_ta_s303	71.86	71.31	40
B_crnn_freq_ca_s101	75.38	74.38	45
B_crnn_freq_ca_s202	74.62	73.64	58
B_crnn_freq_ca_s303	73.12	72.41	57
B_crnn_freq_tca_s101	74.12	72.86	28
B_crnn_freq_tca_s202	71.61	70.21	58
B_crnn_freq_tca_s303	72.86	72.02	40
C_specaugment_s101	72.86	71.96	45
C_specaugment_s202	70.60	69.25	58
C_specaugment_s303	72.11	71.04	38
C_noise_s101	72.61	71.28	57
C_noise_s202	74.12	73.41	41
C_noise_s303	72.61	71.29	52
C_mixup_s101	72.11	71.16	58
C_mixup_s202	70.85	70.40	56
C_mixup_s303	71.86	70.41	57
D_sr16_fmax8_logmel64_s101	49.25	46.02	59
D_sr16_fmax8_logmel64_s202	52.51	49.08	49
D_sr16_fmax8_logmel64_s303	53.02	50.27	57
D_sr16_fmax8_logmel32_s101	54.77	52.42	58
D_sr16_fmax8_logmel32_s202	56.03	53.28	58
D_sr16_fmax8_logmel32_s303	56.03	53.21	57
D_sr16_fmax8_mfcc40_s101	48.24	45.71	58

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
D_sr16_fmax8_mfcc40_s202	51.26	49.05	56
D_sr16_fmax8_mfcc40_s303	48.24	46.48	59
D_sr16_fmax8_loglinear64_s101	51.01	47.65	58
D_sr16_fmax8_loglinear64_s202	51.26	48.22	49
D_sr16_fmax8_loglinear64_s303	52.51	49.87	57
D_sr32_fmax8_logmel64_s101	48.99	45.78	59
D_sr32_fmax8_logmel64_s202	52.26	49.10	49
D_sr32_fmax8_logmel64_s303	52.51	49.83	57
D_sr32_fmax8_logmel32_s101	56.53	54.17	58
D_sr32_fmax8_logmel32_s202	54.52	51.54	58
D_sr32_fmax8_logmel32_s303	56.03	53.31	59
D_sr32_fmax8_mfcc40_s101	48.49	46.16	58
D_sr32_fmax8_mfcc40_s202	51.76	49.53	56
D_sr32_fmax8_mfcc40_s303	49.25	47.18	59
D_sr32_fmax8_loglinear64_s101	50.25	46.63	58
D_sr32_fmax8_loglinear64_s202	50.50	46.72	49
D_sr32_fmax8_loglinear64_s303	51.51	49.39	57
D_sr32_fmax16_logmel64_s101	50.75	47.91	58
D_sr32_fmax16_logmel64_s202	51.51	48.84	49
D_sr32_fmax16_logmel64_s303	51.01	48.23	57
D_sr32_fmax16_logmel32_s101	56.78	54.24	58
D_sr32_fmax16_logmel32_s202	56.28	53.28	56
D_sr32_fmax16_logmel32_s303	58.29	56.26	57
D_sr32_fmax16_mfcc40_s101	46.98	44.76	54
D_sr32_fmax16_mfcc40_s202	53.02	50.50	56
D_sr32_fmax16_mfcc40_s303	48.74	46.62	59
D_sr32_fmax16_loglinear64_s101	47.99	45.38	58
D_sr32_fmax16_loglinear64_s202	48.49	44.81	49
D_sr32_fmax16_loglinear64_s303	50.50	48.07	57

Appendix D. Completed three-seed robustness ledger

All rows use the same 398 validation recordings. Training seeds identify fixed fits, not independent audio datasets.

Model	Seed	Condition	Accuracy (%)	Correct	Clips
cnn	101	clean	72.61	289	398
cnn	101	snr_20	47.74	190	398
cnn	101	snr_10	23.37	93	398
cnn	101	snr_0	6.78	27	398
cnn	101	gain_-6	68.84	274	398
cnn	101	gain_6	71.86	286	398
cnn	202	clean	72.36	288	398
cnn	202	snr_20	49.25	196	398
cnn	202	snr_10	24.37	97	398
cnn	202	snr_0	7.54	30	398
cnn	202	gain_-6	69.85	278	398
cnn	202	gain_6	70.60	281	398
cnn	303	clean	72.86	290	398
cnn	303	snr_20	46.23	184	398
cnn	303	snr_10	22.36	89	398
cnn	303	snr_0	7.29	29	398
cnn	303	gain_-6	70.10	279	398
cnn	303	gain_6	71.86	286	398
crnn	101	clean	75.13	299	398
crnn	101	snr_20	54.77	218	398
crnn	101	snr_10	24.62	98	398
crnn	101	snr_0	6.28	25	398
crnn	101	gain_-6	69.85	278	398
crnn	101	gain_6	73.37	292	398
crnn	202	clean	73.87	294	398
crnn	202	snr_20	48.24	192	398
crnn	202	snr_10	23.12	92	398
crnn	202	snr_0	5.53	22	398

Model	Seed	Condition	Accuracy (%)	Correct	Clips
crnn	202	gain_-6	69.35	276	398
crnn	202	gain_6	70.60	281	398
crnn	303	clean	72.61	289	398
crnn	303	snr_20	49.50	197	398
crnn	303	snr_10	23.87	95	398
crnn	303	snr_0	5.78	23	398
crnn	303	gain_-6	69.10	275	398
crnn	303	gain_6	70.60	281	398
noise_cnn	101	clean	72.61	289	398
noise_cnn	101	snr_20	70.60	281	398
noise_cnn	101	snr_10	62.81	250	398
noise_cnn	101	snr_0	26.13	104	398
noise_cnn	101	gain_-6	68.59	273	398
noise_cnn	101	gain_6	70.60	281	398
noise_cnn	202	clean	74.12	295	398
noise_cnn	202	snr_20	70.10	279	398
noise_cnn	202	snr_10	58.79	234	398
noise_cnn	202	snr_0	21.86	87	398
noise_cnn	202	gain_-6	68.59	273	398
noise_cnn	202	gain_6	70.10	279	398
noise_cnn	303	clean	72.61	289	398
noise_cnn	303	snr_20	70.10	279	398
noise_cnn	303	snr_10	62.31	248	398
noise_cnn	303	snr_0	28.14	112	398
noise_cnn	303	gain_-6	68.59	273	398
noise_cnn	303	gain_6	72.11	287	398

Appendix E. Temporal-intervention ledger

Changed predictions are relative to each checkpoint's own identity predictions. The intervention reorders encoded tokens, not waveform samples. The first float32 absolute-tolerance failure and corrected numerical interpretation are retained in the project records.

Model	Seed	Condition	Accuracy (%)	Changed predictions	Clips
cnn_freq	101	identity	72.61	0	398
cnn_freq	101	reverse	72.61	0	398
cnn_freq	101	roll19	72.61	0	398
cnn_freq	101	shuffle17	72.61	0	398
cnn_freq	101	shuffle29	72.61	0	398
cnn_freq	202	identity	72.36	0	398
cnn_freq	202	reverse	72.36	0	398
cnn_freq	202	roll19	72.36	0	398
cnn_freq	202	shuffle17	72.36	0	398
cnn_freq	202	shuffle29	72.36	0	398
cnn_freq	303	identity	72.86	0	398
cnn_freq	303	reverse	72.86	0	398
cnn_freq	303	roll19	72.86	0	398
cnn_freq	303	shuffle17	72.86	0	398
cnn_freq	303	shuffle29	72.86	0	398
crnn_freq	101	identity	75.13	0	398
crnn_freq	101	reverse	69.10	71	398
crnn_freq	101	roll19	73.37	39	398
crnn_freq	101	shuffle17	72.86	39	398
crnn_freq	101	shuffle29	74.12	37	398
crnn_freq	202	identity	73.87	0	398
crnn_freq	202	reverse	68.84	78	398
crnn_freq	202	roll19	73.62	38	398
crnn_freq	202	shuffle17	73.62	39	398
crnn_freq	202	shuffle29	74.62	41	398
crnn_freq	303	identity	72.61	0	398
crnn_freq	303	reverse	68.59	57	398
crnn_freq	303	roll19	70.35	35	398
crnn_freq	303	shuffle17	71.86	39	398
crnn_freq	303	shuffle29	72.61	34	398

Appendix F. Sources, analysis and completion status

• Official ESC-50 dataset and credits. Dataset license and recording metadata remain in the project.

• Original matrix summary tables [P01].

• Completed prediction and robustness analysis [P02].

• Temporal-order probe and numerical implementation record [P03].

• New attention paired effects [P04].

• Model-decision meeting and manuscript review [P05].

• Initial attention methods, with completed-training update [P06].

• PANNs author repository and AST author repository document external-pretraining routes. No such models were downloaded, trained or evaluated here; published benchmark protocols and extra data differ from this study.

This report closes the current executed research scope. It preserves the original study and all available figure originals. Further model-performance optimization is a separately proposed phase, not an unperformed requirement hidden inside this completion claim. The report and computational analyses are agent-produced; role-based ChatGPT discussion is not independent human peer review. No new training, test-fold inference, file-hash verification or test-suite expansion was performed to produce this final document.

Project evidence index

The paths below refer to the accompanying ESC-50 project. They remain useful provenance references when the PDF is shared, but the project files are not embedded in this PDF. Public web references in the report are clickable.

[P01] `reports/evidence/matrix_20261028/analysis`

[P02] `reports/followup_20261104/report_en.md`

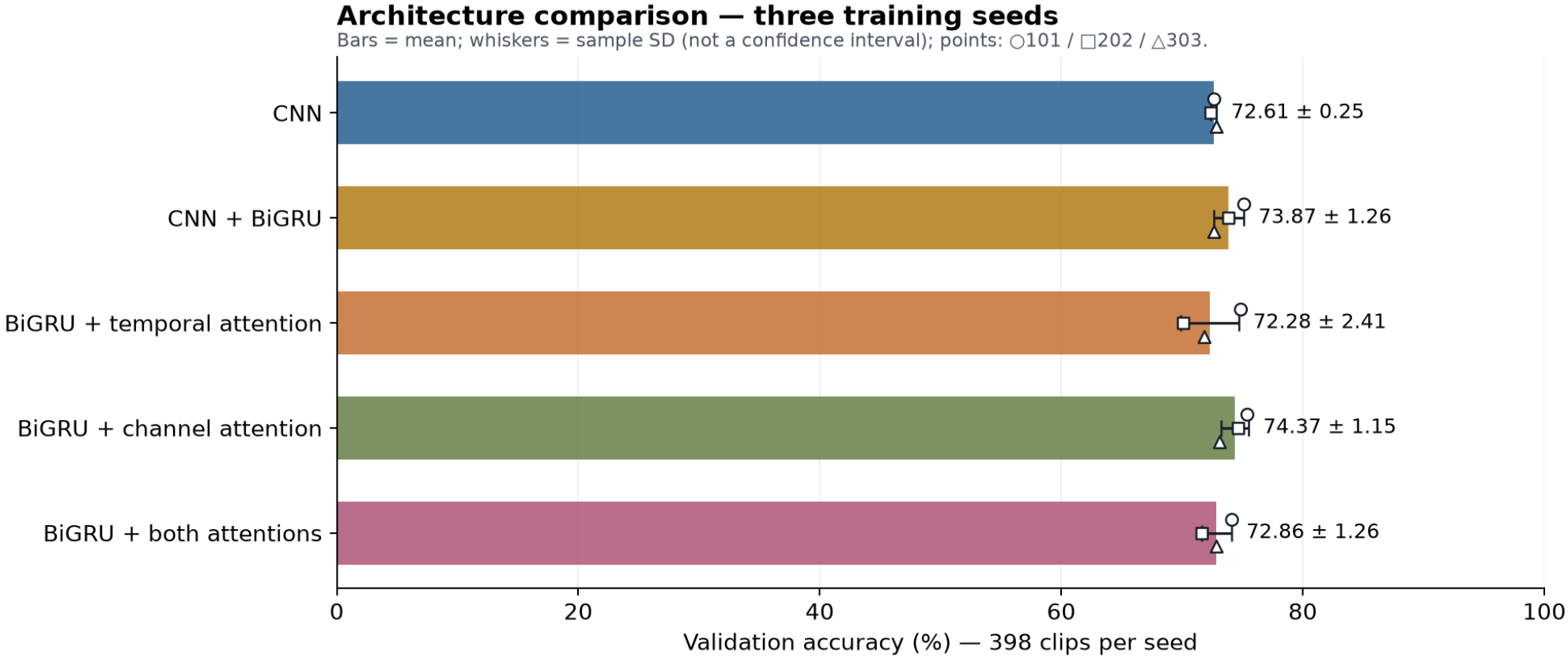
[P03] `reports/temporal_20261105/README.md`

[P04] `reports/final_20261105/attention_paired_effects.json`

[P05] `docs/research_meetings/2026-11-05_model_decision.md`

[P06] `docs/methodology/attention_phase1.md`

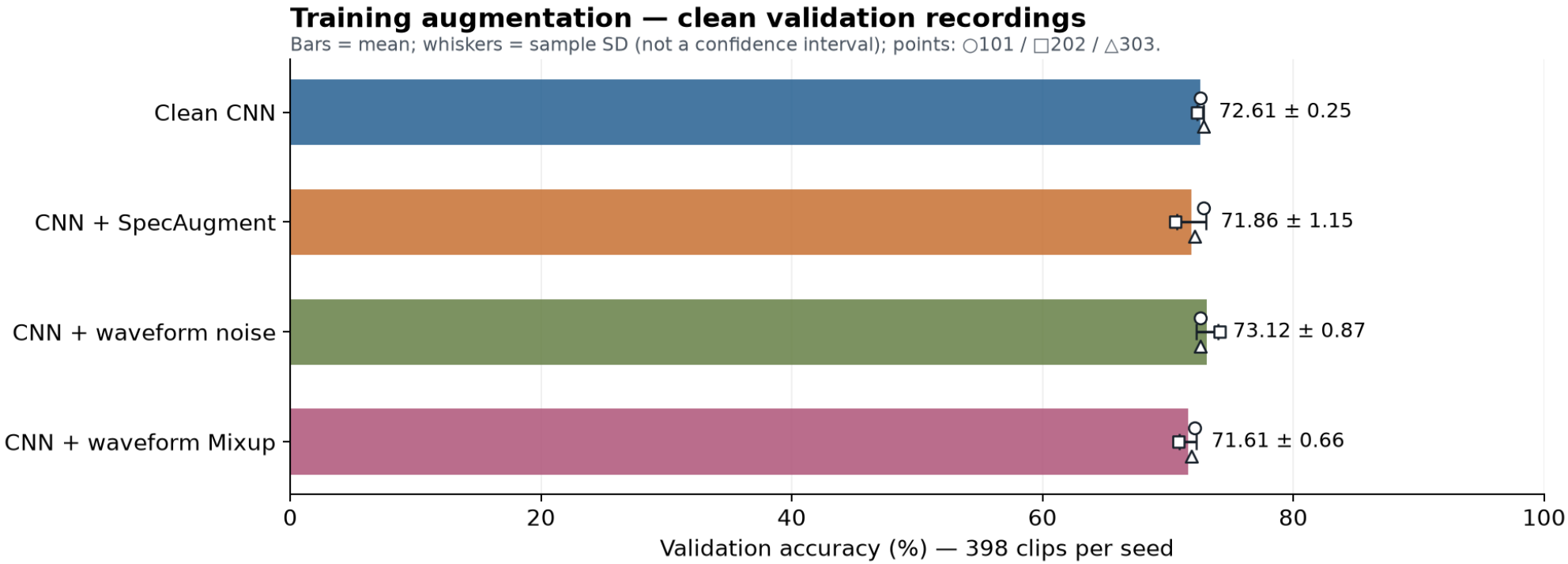
Figure 1. Three-seed development validation; SD is not a confidence interval.



Three-seed development validation; SD is not a confidence interval.

Source: `reports/figures/matrix_20261028/architecture.png`. Original saved figure; no experiment was rerun for this PDF.

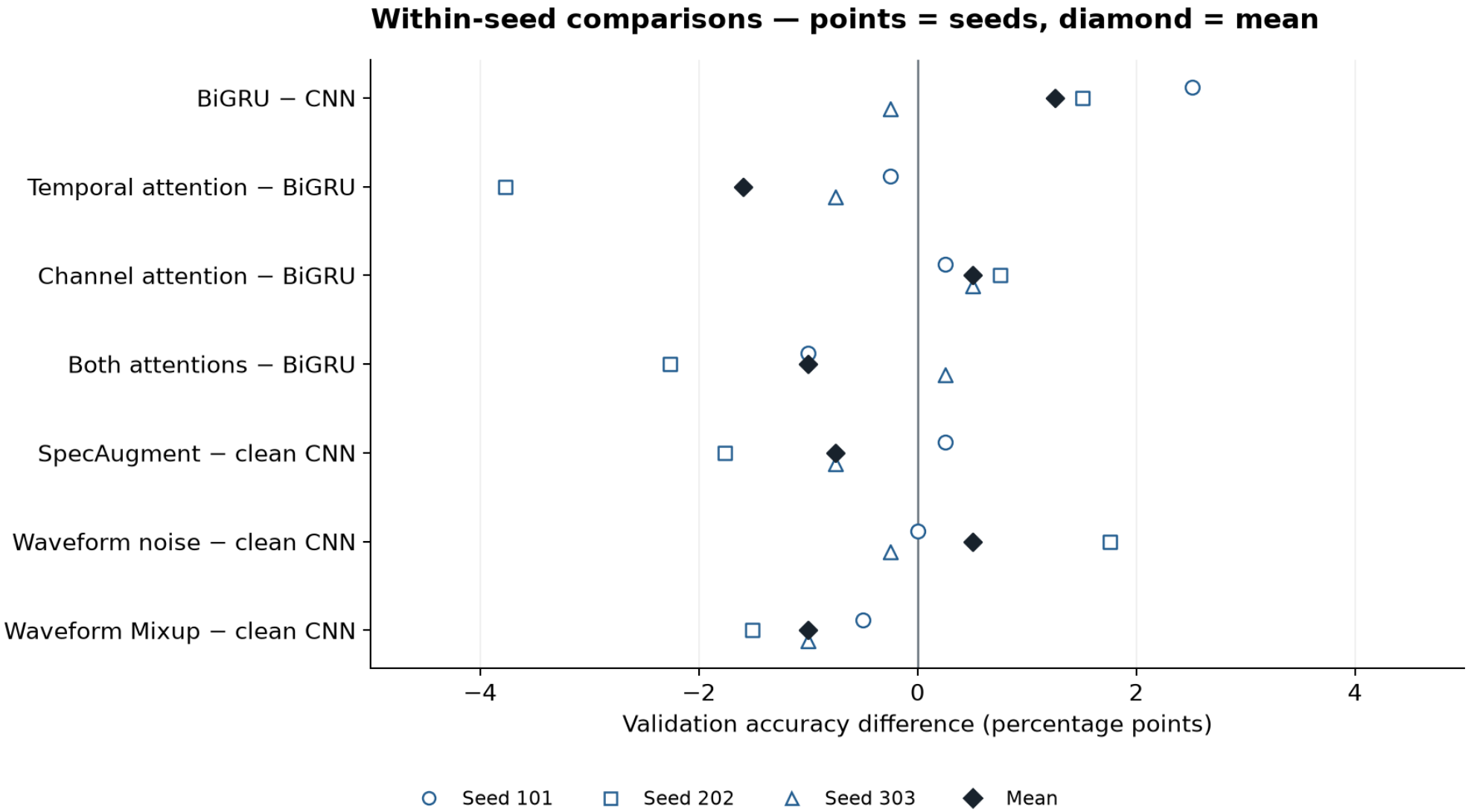
Figure 2. Augmentation affects training only; validation recordings stay clean.



Augmentation affects training only; validation recordings stay clean.

Source: `reports/figures/matrix_20261028/augmentation.png`. Original saved figure; no experiment was rerun for this PDF.

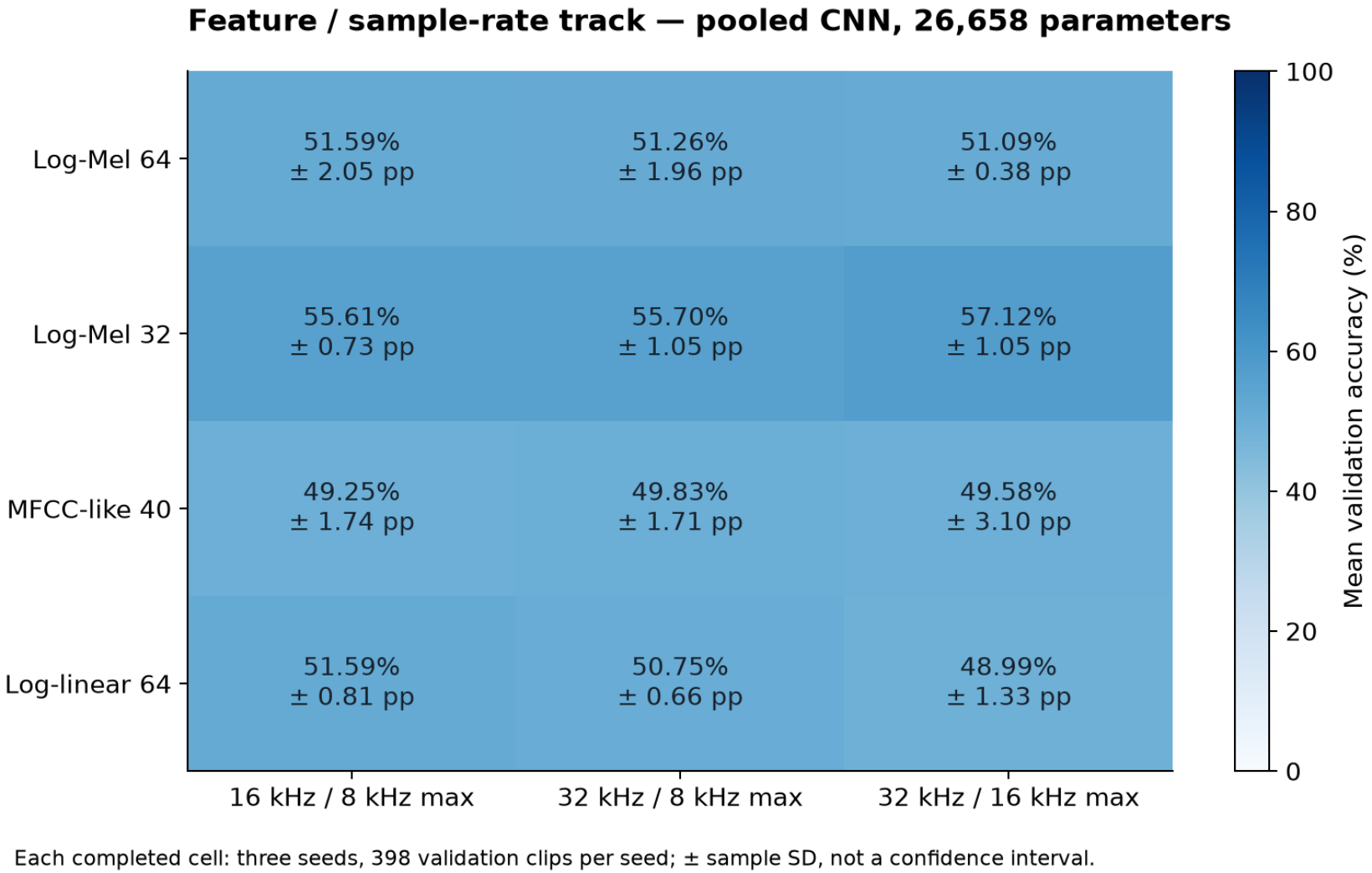
Figure 3. Within-seed deltas; these are not significance tests.



Within-seed deltas; these are not significance tests.

Source: `reports/figures/matrix_20261028/paired_effects.png`. Original saved figure; no experiment was rerun for this PDF.

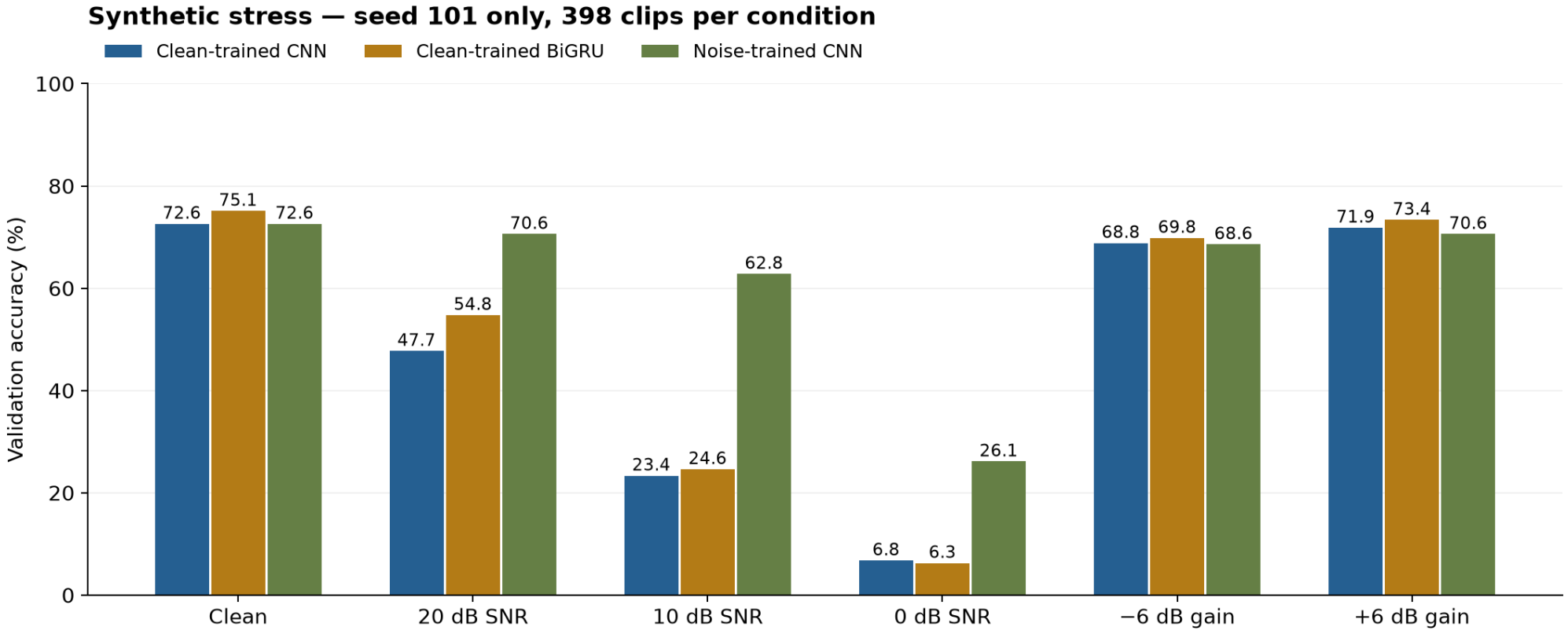
Figure 4. Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track.



Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track.

Source: `reports/figures/matrix_20261028/feature_sampling.png`. Original saved figure; no experiment was rerun for this PDF.

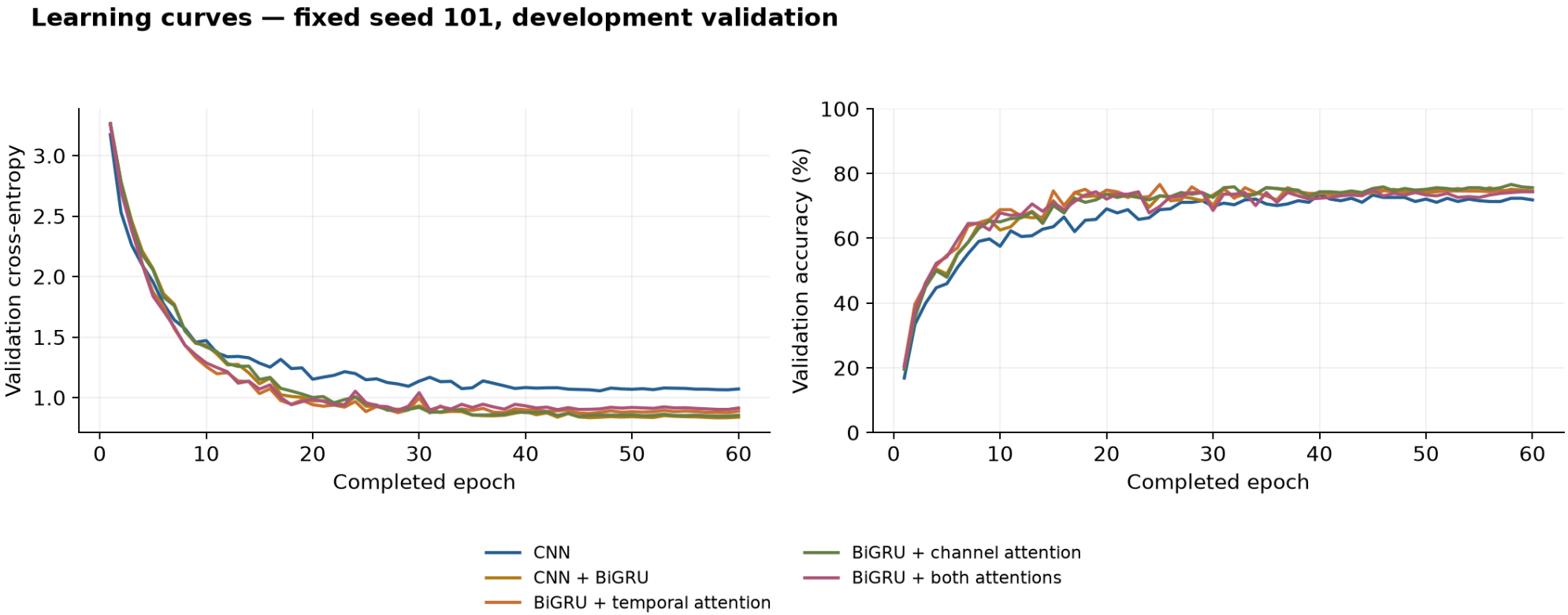
Figure 5. Original seed101 robustness comparison: one fitted checkpoint per model



Original seed101 robustness comparison: one fitted checkpoint per model

Source: `reports/figures/matrix_20261028/robustness.png`. Original saved figure; no experiment was rerun for this PDF.

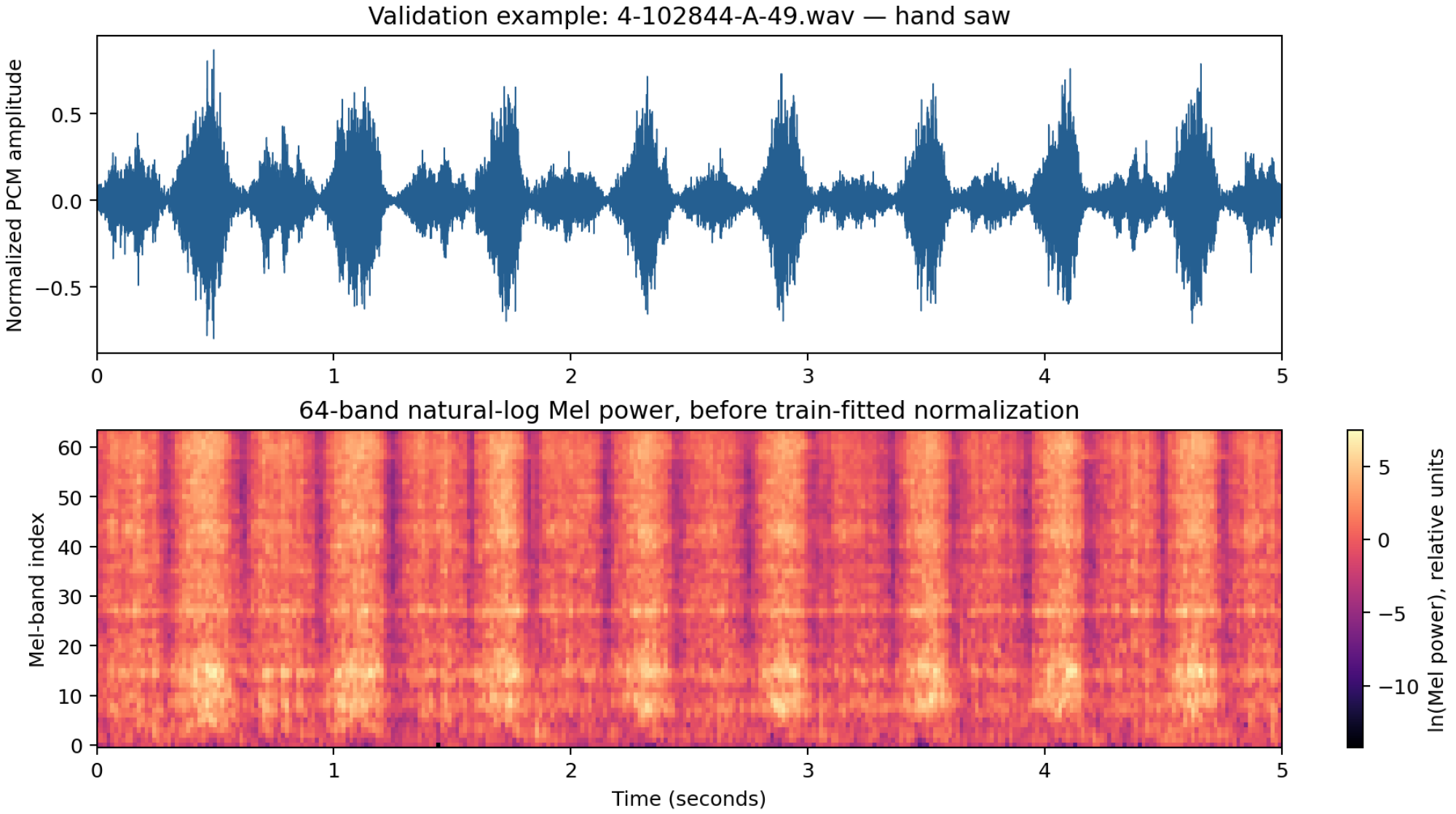
Figure 6. Fixed-seed101 validation curves; checkpoints use minimum cross-entropy.



Fixed-seed101 validation curves; checkpoints use minimum cross-entropy.

Source: `reports/figures/matrix_20261028/learning_curves.png`. Original saved figure; no experiment was rerun for this PDF.

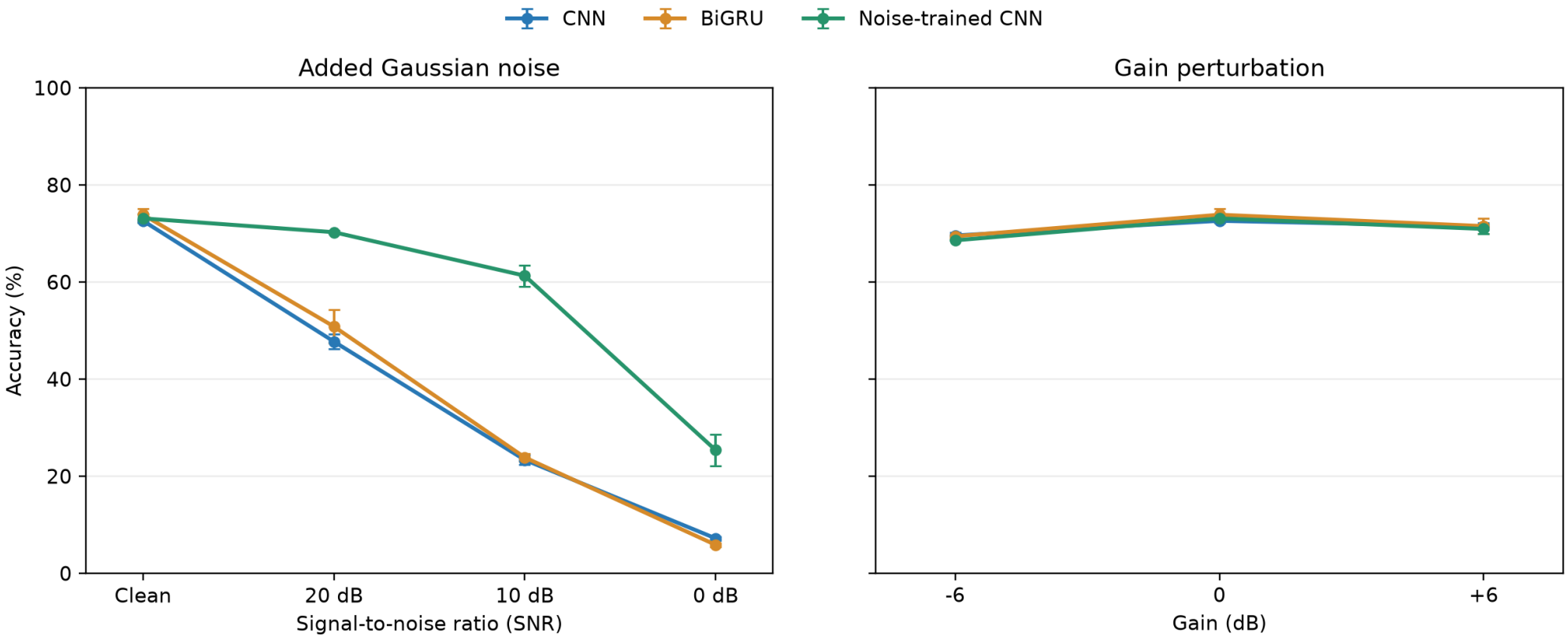
Figure 7. A real validation waveform and Log-Mel representation



A real validation waveform and Log-Mel representation

Source: `reports/figures/matrix_20261028/frontend_example.png`. Original saved figure; no experiment was rerun for this PDF.

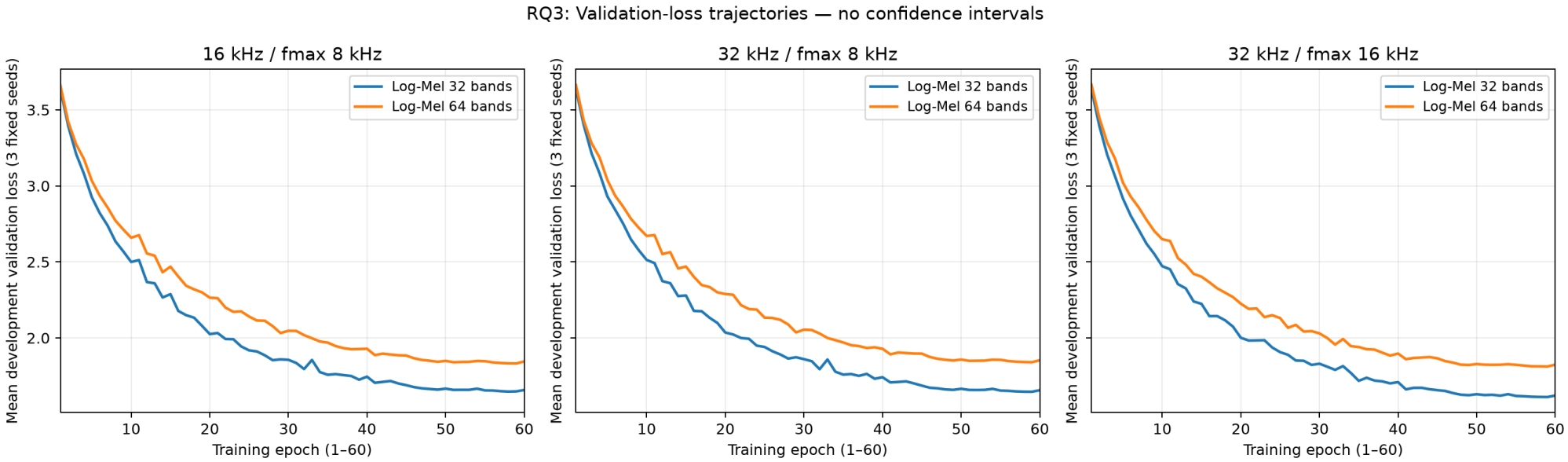
Figure 8. Completed three-seed Gaussian-noise and gain results; bars show sample SD, not confidence intervals



Completed three-seed Gaussian-noise and gain results; bars show sample SD, not confidence intervals

Source: `reports/final_20261105/three_seed_robustness.png`. Original saved figure; no experiment was rerun for this PDF.

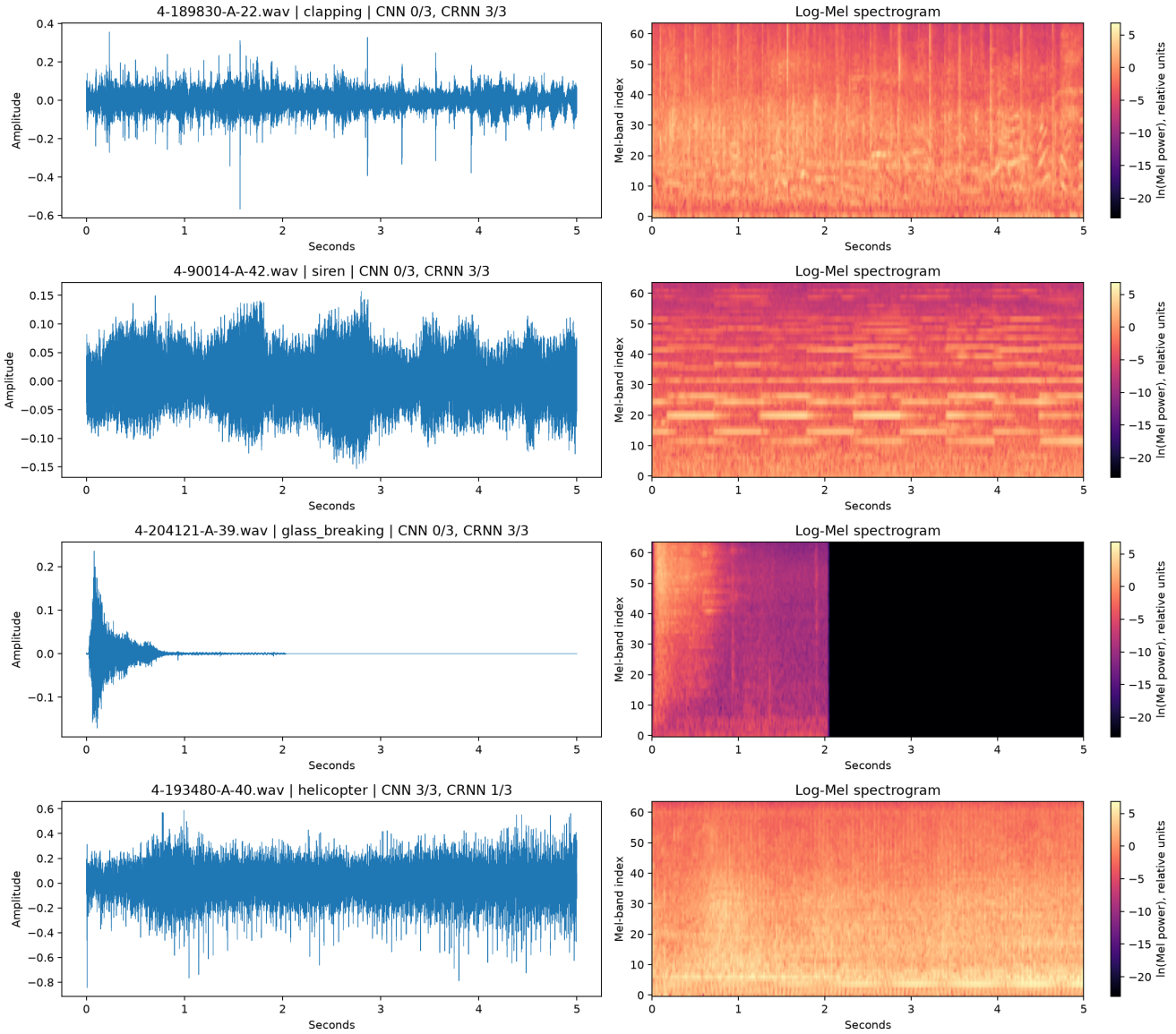
Figure 9. Feature-study loss trajectories: Log-Mel 32 and Log-Mel 64, three fixed seeds



Feature-study loss trajectories: Log-Mel 32 and Log-Mel 64, three fixed seeds

Source: `reports/followup_20261104/convergence/validation_loss_curves.png`. Original saved figure; no experiment was rerun for this PDF.

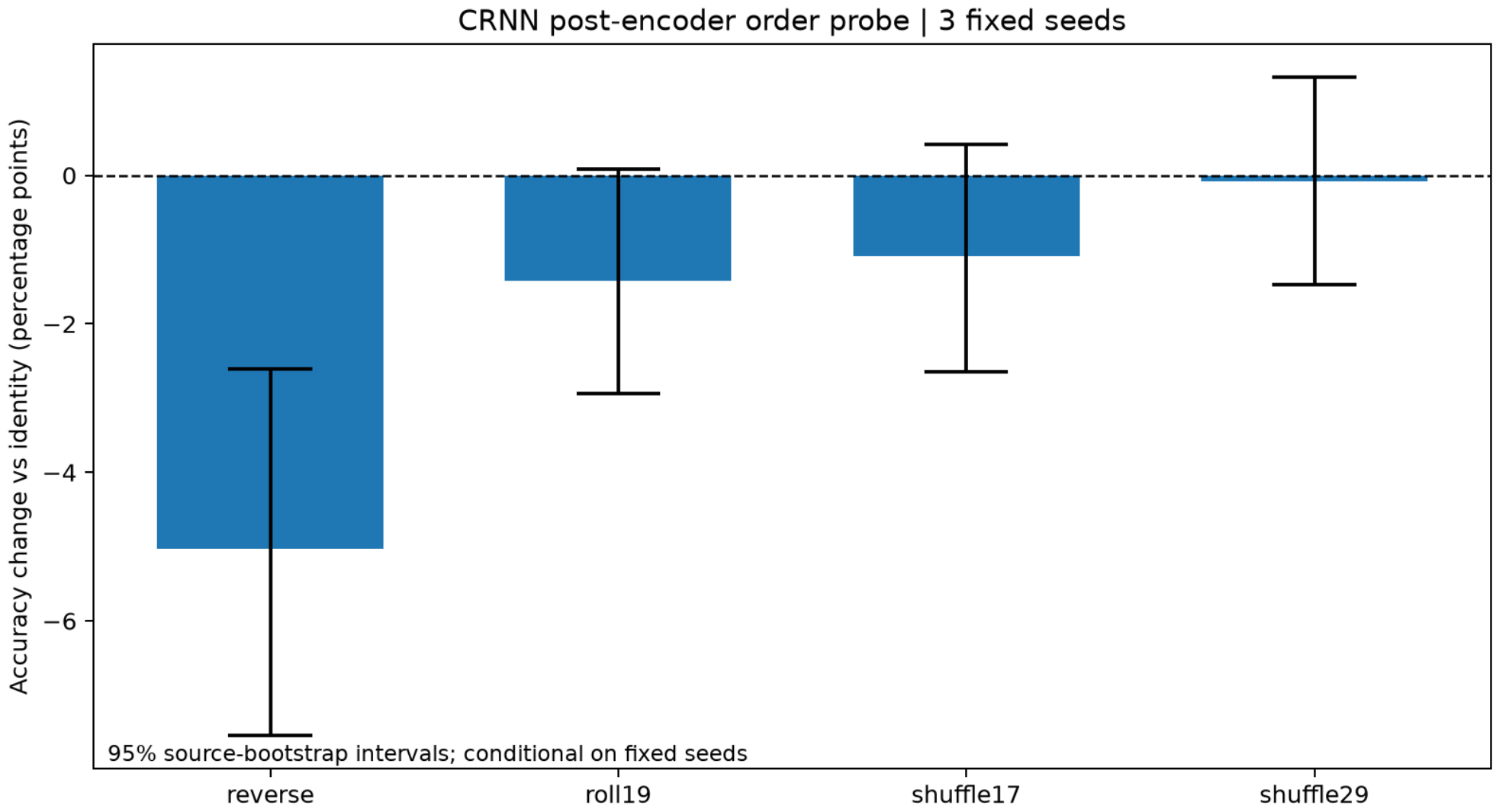
Figure 10. Four selected real-audio error cases: waveform and Log-Mel evidence, not causal explanations



Four selected real-audio error cases: waveform and Log-Mel evidence, not causal explanations

Source: `reports/followup_20261104/error_cases/error_case_atlas.png`. Original saved figure; no experiment was rerun for this PDF.

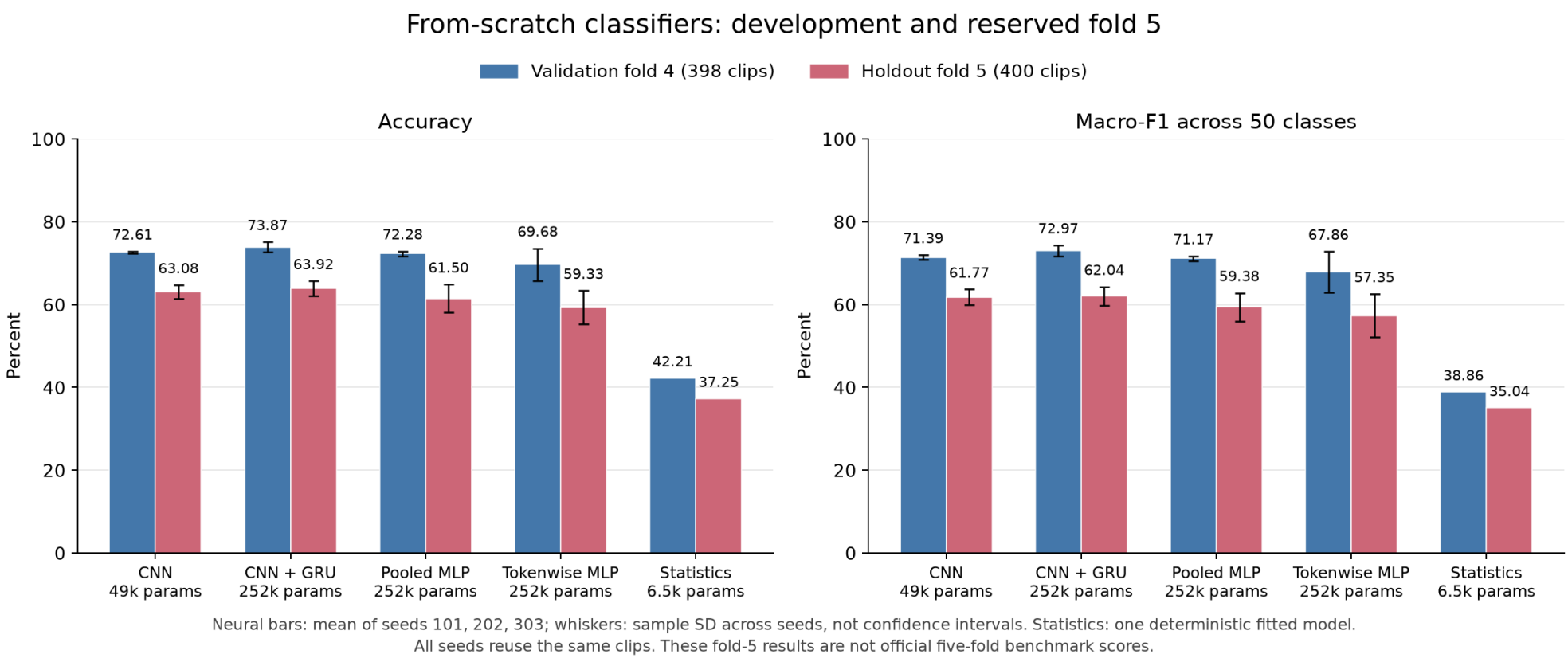
Figure 11. CRNN encoder-token intervention effects with conditional source-bootstrap intervals



CRNN encoder-token intervention effects with conditional source-bootstrap intervals

Source: `reports/temporal_20261105/crnn_accuracy_change.png`. Original saved figure; no experiment was rerun for this PDF.

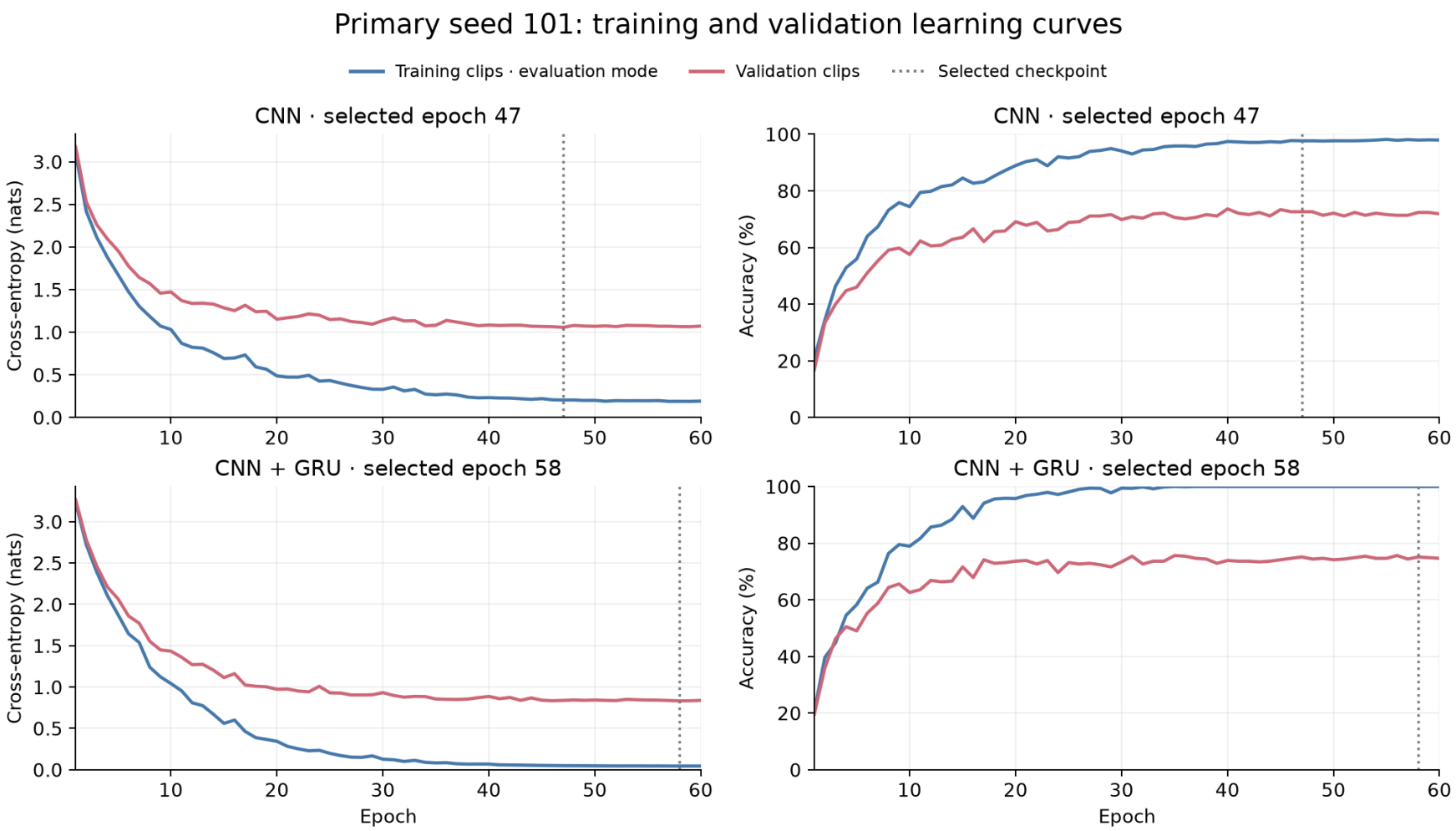
Figure 12. Historical study - Model comparison



Historical study - Model comparison

Source: `reports/figures/baseline_20261028/model_comparison.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 13. Historical study - Learning curves

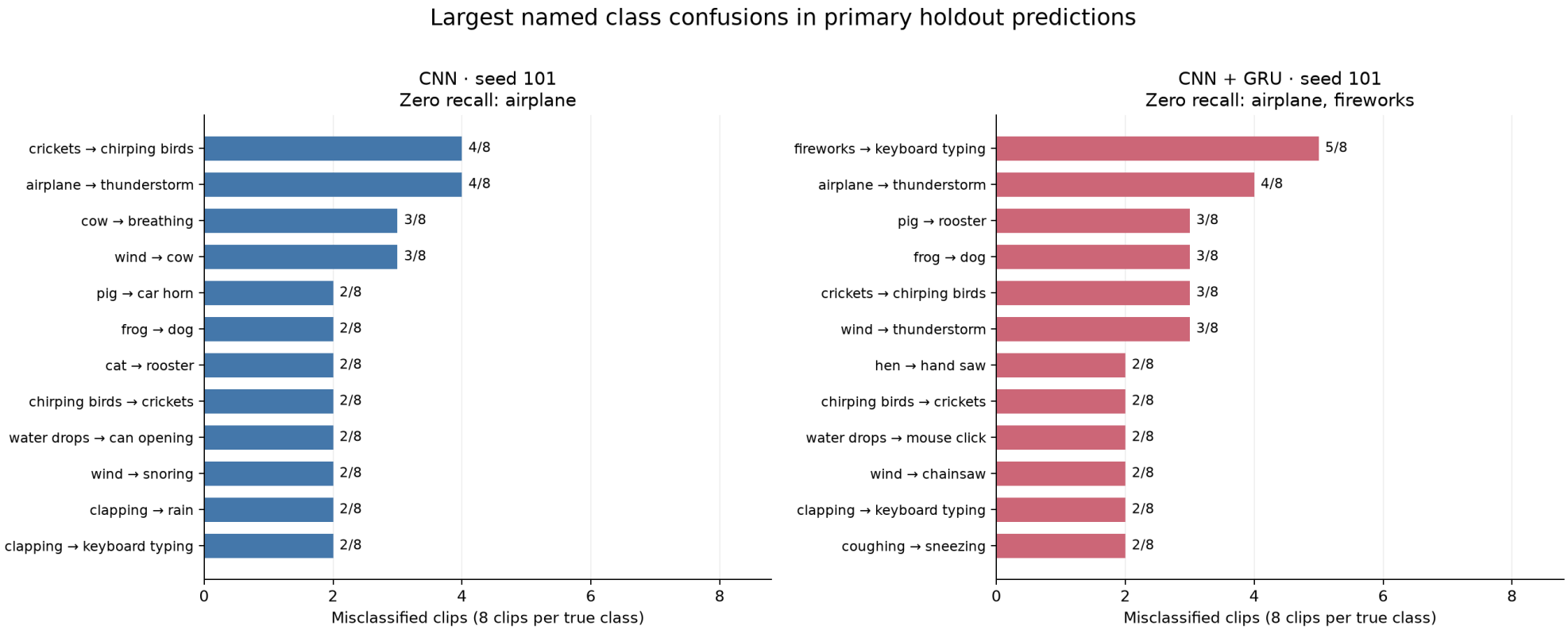


Both curves are measured at epoch end in evaluation mode; online optimization loss is excluded. Train: 1,200 clips; validation: 398 clips.
Checkpoint rule: earliest minimum validation cross-entropy. These are development curves for one seed, not holdout learning curves.

Historical study - Learning curves

Source: `reports/figures/baseline_20261028/learning_curves.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 14. Historical study - Named confusions

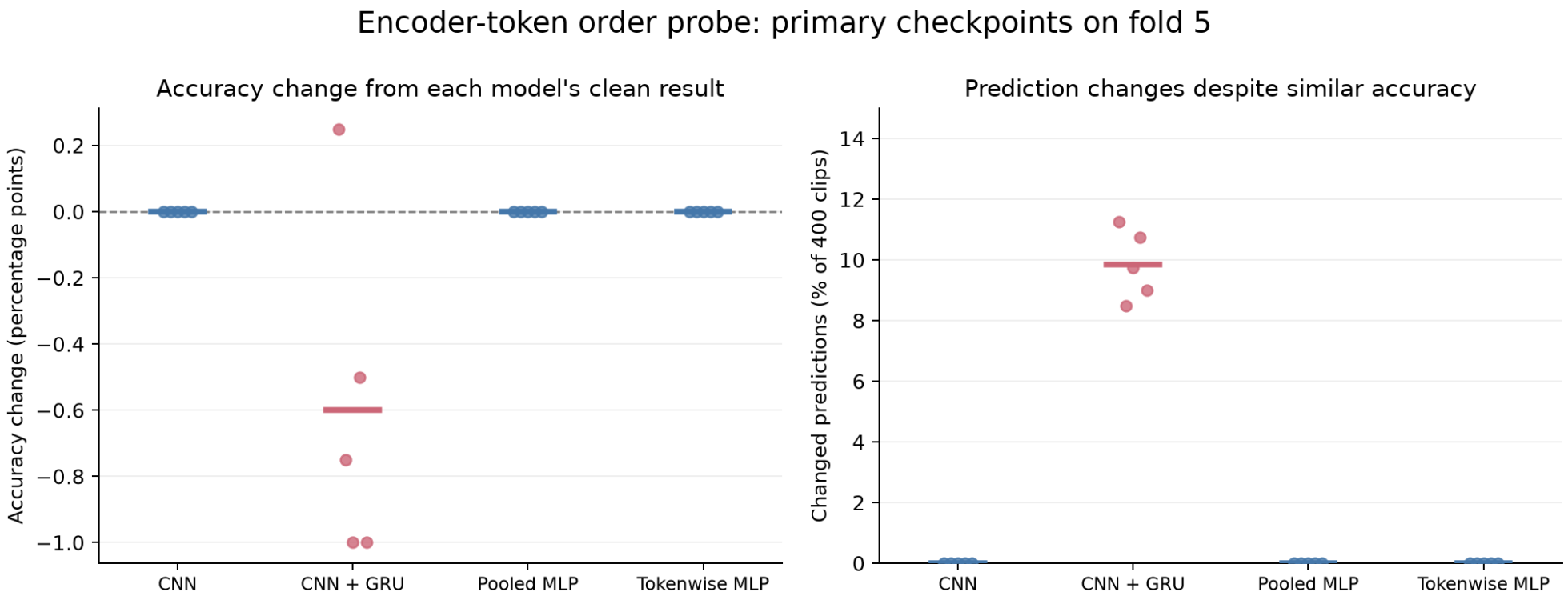


Largest off-diagonal counts, described after evaluation; they were not used to retune the frozen models. A true class may appear in more than one row.
Eight clips per class give coarse estimates. These errors do not characterize every real-world recording condition.

Historical study - Named confusions

Source: `reports/figures/baseline_20261028/top_confusions.png`. Original saved figure; no experiment was rerun for this PDF.

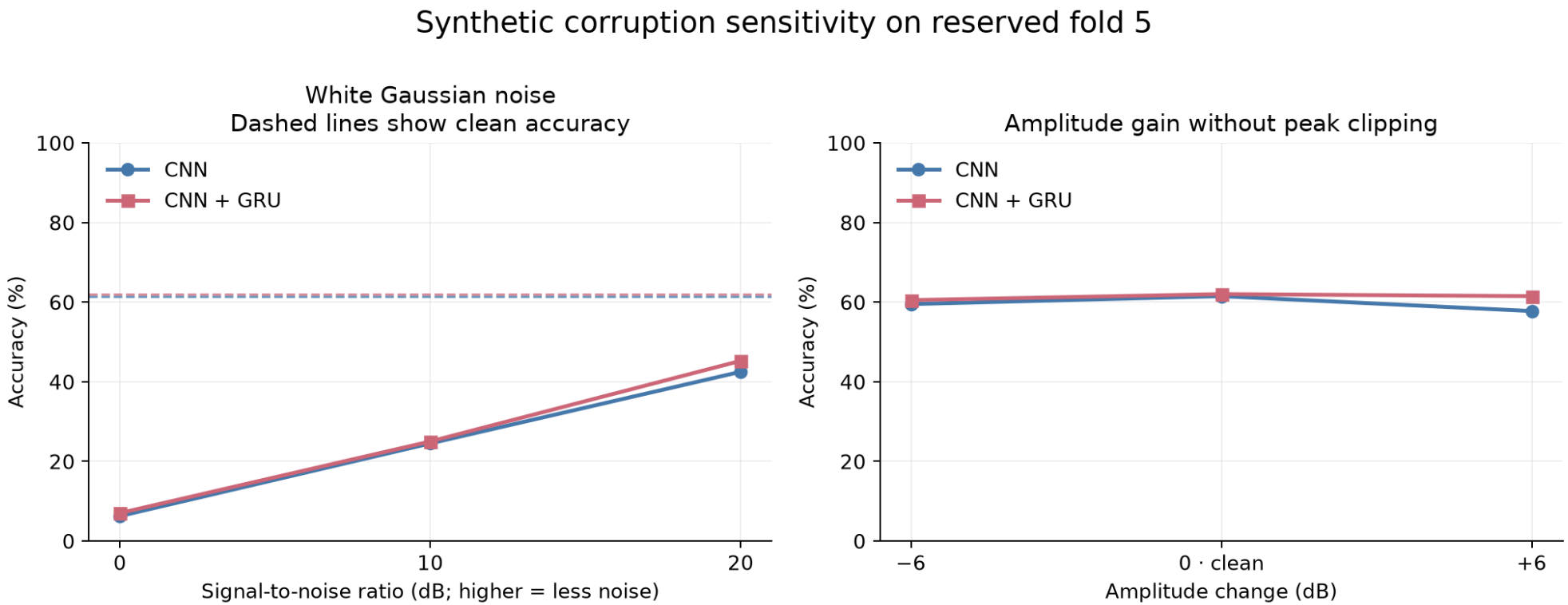
Figure 15. Historical study - Order probe



Historical study - Order probe

Source: `reports/figures/baseline_20261028/order_probe.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 16. Historical study - Synthetic stress

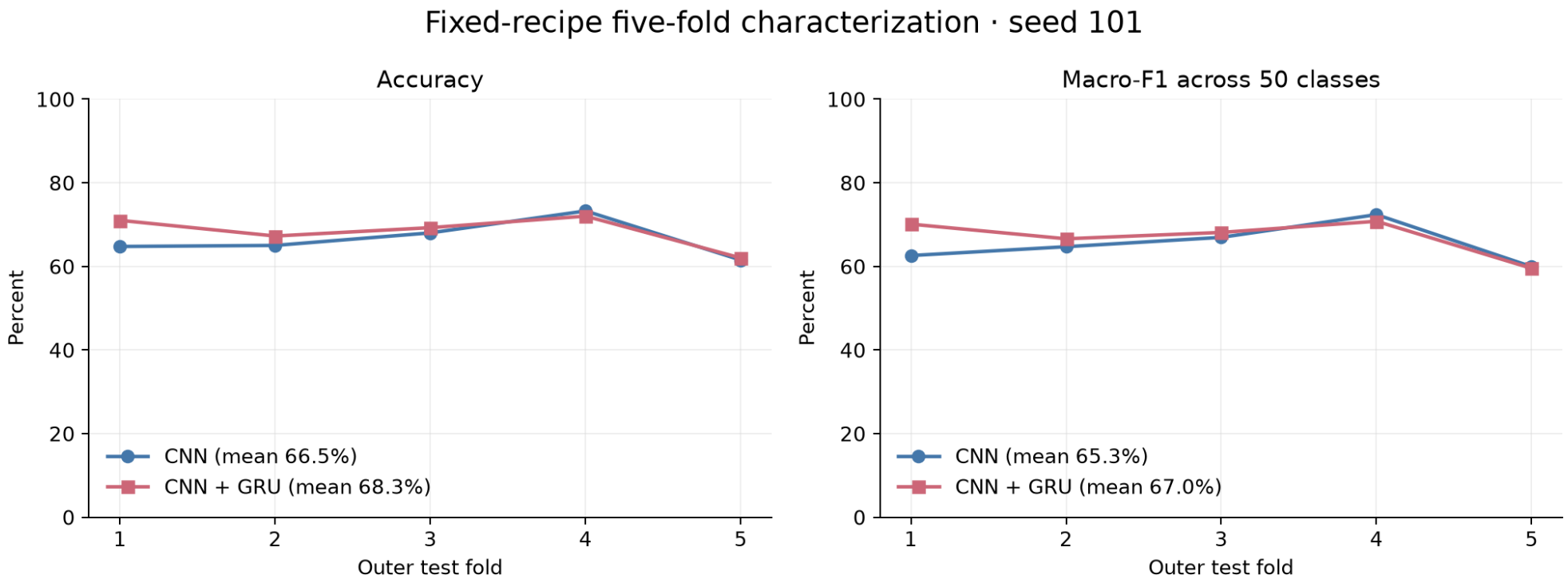


Seed 101 only; same 400 holdout clips. One deterministic noise realization per clip/condition (noise seed 2026); no uncertainty bars.
Synthetic corruption sensitivity does not establish real-world acoustic robustness. Clean reference: CNN 61.5%, CNN + GRU 62.0%.

Historical study - Synthetic stress

Source: `reports/figures/baseline_20261028/synthetic_stress.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 17. Historical study - Five-fold characterization

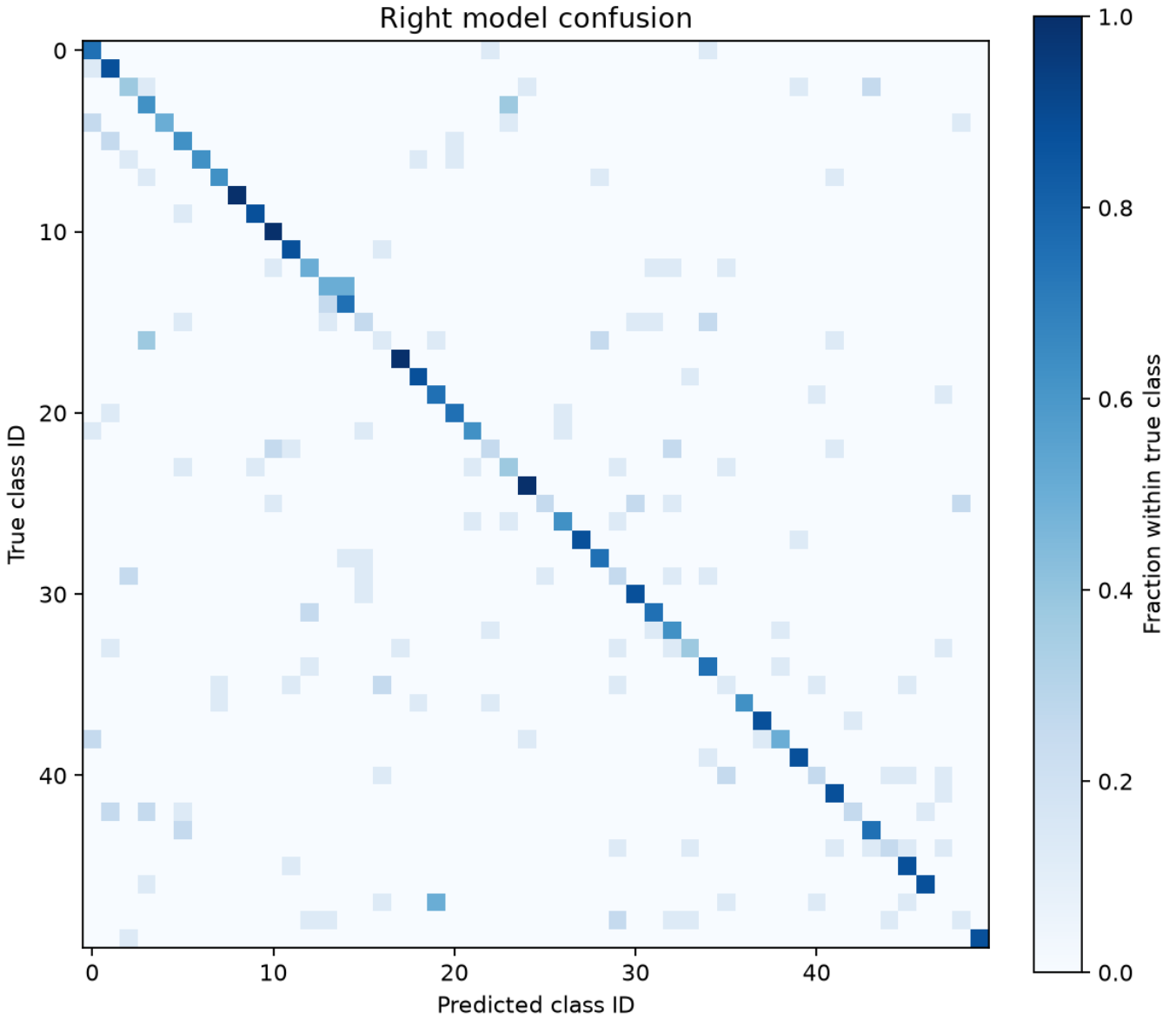


Source-strict rotating folds; one seed. Folds 1–4 already informed development: this is not nested model selection or five untouched confirmations. Fold scores share training data and are dependent; no confidence intervals are shown.

Historical study - Five-fold characterization

Source: `reports/figures/baseline_20261028/cross_validation.png`. Original saved figure; no experiment was rerun for this PDF.

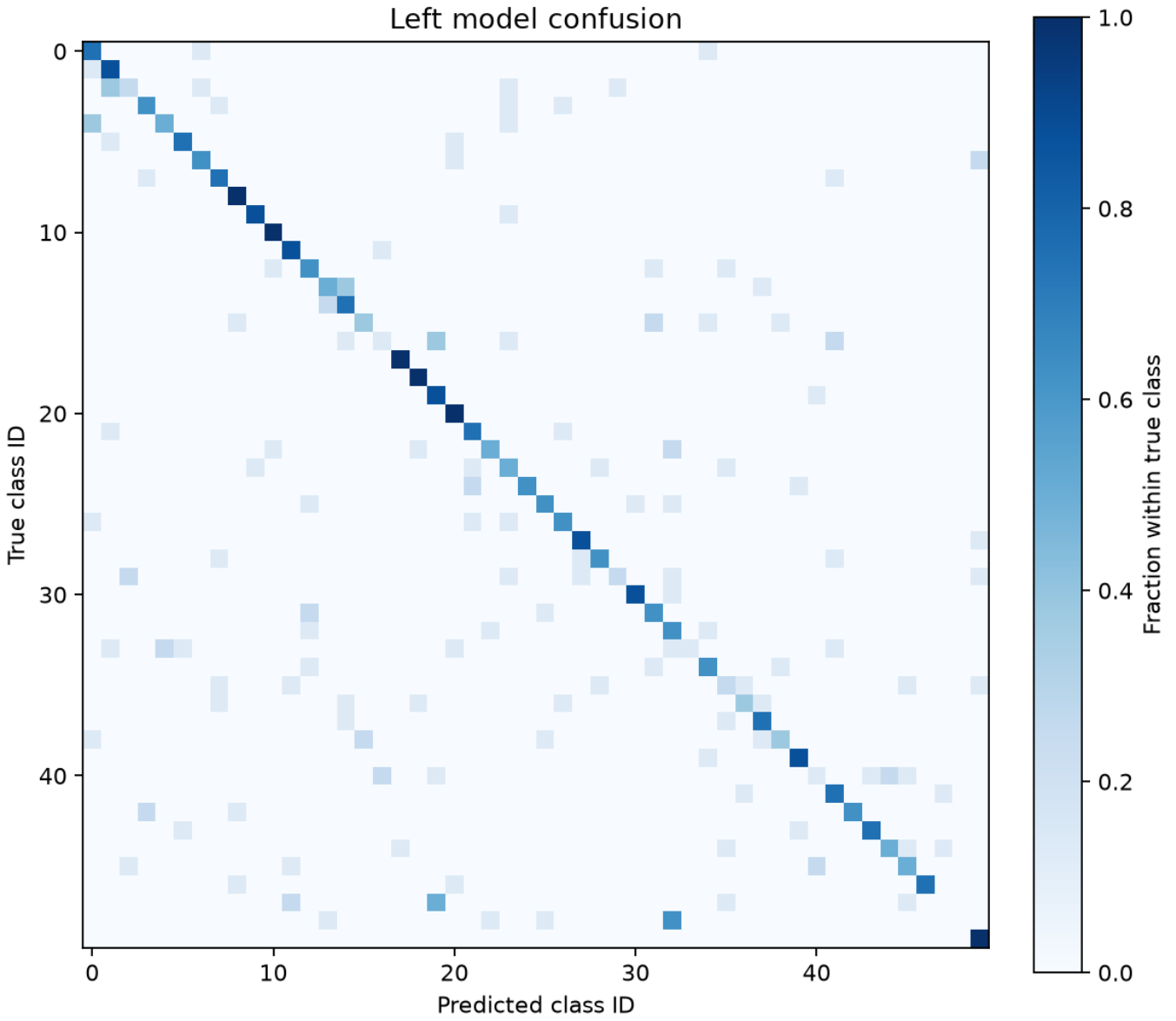
Figure 18. Historical study - CNN primary fold-5 confusion matrix



Historical study - CNN primary fold-5 confusion matrix

Source: `reports/figures/baseline_20261028/cnn_confusion.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 19. Historical study - CRNN (CNN + BiGRU) primary fold-5 confusion matrix



Historical study - CRNN (CNN + BiGRU) primary fold-5 confusion matrix

Source: `reports/figures/baseline_20261028/crnn_confusion.png`. Original saved figure; no experiment was rerun for this PDF.